Rafael Stavros **Cottom**

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Ph.D. Student, Fellow

Rm. 203, Wright-Rieman Laboratories, Busch Campus, Rutgers University

Interests

Research | Continuum, fluid, and thermal physics of ice sheets and icy satellite interiors, terrestrial planets, and exoplanets; the evolution of surfaces, ice shells and oceans, and interiors; influences of geophysical and astrophysical processes and events on habitability.

Service | Geoscience and astronomy community outreach; mentoring students.

Education

2030 | **Rutgers University,** New Brunswick, NJ
Ph.D., Earth and Planetary Sciences
Advisor: Lujendra Ojha

2025 | **University of California, Santa Cruz,** Santa Cruz, CA
B.S., Earth Sciences (Planetary Science)
Stressing Out: Episodic Tidal Amplifications as a Mechanism for Band Formation on Europa
Thesis Advisors: Benjamin Idini & Jonathan Fortney

2023 | **Santa Barbara City College,** Santa Barbara, CA
A.S., Astronomy (transfer w/o degree)

Additional Training

2026 | International Heat Flow Commission (IHFC) Summer School | *İzmir, Türkiye*

Research Experience

Aug 2024 - June 2025 | **Undergraduate Researcher**
Solar System Physics (S^2P) Group, Astronomy & Astrophysics Department, UCSC
- Developed a simple thermal model to examine the effect of cooling on Europa's ice shell
- Calculated ice shell transient radial and tangential stress profiles (ongoing)

June 2023 - Aug 2023 | **Cosmology Research Intern**
McGill Cosmic Dawn and Hydrogen Cosmology Group
- Modeled the distribution of neutral H in the Epoch of Reionization

June 2022 - Aug 2022 | **Astrophysics Research Intern**
Carnegie Observatories — Carnegie Astrophysics Summer Student Internship (CASSI)
- Derived the IMF for a group of metal-poor stars by matching chemical abundance patterns with SNe models

Publications

in prep | **Cottom, R.** & Holmbeck, E. (2026). Deriving the First Stellar Mass Function from the Light-Element Abundances of Metal-poor Stars.

Honors & Grants

2025 | Dean's Honors List (Spring Quarter) · *University of California, Santa Cruz*
Excellence Fellowship · *Earth & Planetary Sciences, Rutgers University* ($40,000 USD)
Eli Silver Earth & Planetary Science Opportunities Fund · *Earth & Planetary Sciences, UCSC* ($2,000 USD)
2023 | Marschak Astronomy SBCC Foundation Scholarship · *Earth & Planetary Sciences, SBCC* ($1,000 USD)
2022 | Neil Currie Award: Outstanding Planetary Petrologist · *Earth & Planetary Sciences, SBCC* ($500 USD)
Total: | $43,500 USD

Professional Experience

2019-2023 Lead Astronomy Programs Presenter, Coordinator · *Astronomy Department & School and Teacher Services*, Santa Barbara Natural History Museum (SBMNH)

2023 Tutor · *Field Course: Death Valley Petrology (5 Days and 4 Nights)*, Earth and Planetary Sciences, SBCC

Conference Presentations

Posters

2026 **Cottom, R.** & Ojha, L. (2026). Dike-Mediated Water Transport in the Europan Ice Shell: Implications for the Feasibility of Ice-Ocean Exchange. *57th Lunar and Planetary Science Conference*, #1881.

2023 **Cottom, R.** & Holmbeck, E. (2023). Deriving the First Stellar Mass Function from the Light-Element Abundances of Metal-poor Stars. *Bulletin of the American Astronomical Society, 55*(2).

Other Presentations

2025 *Ocean Worlds and Planetary Science* · **Invited Speaker**, SBCC Astronomy Club

2023 *Algal Members of the Crystal Springs Formation* · **Tutor Talk**, Death Valley Petrology Field Course, SBCC

Galactic Archaeology · **Invited Speaker**, Santa Barbara Astronomical Unit: First Friday of the Month Speaker Event

Astronomy 101 · **Speaker**, Educator Training, Santa Barbara Museum of Natural History

2022 *Galactic Archaeology* · **Invited Speaker**, Science Pub at Dargan's, Santa Barbara Museum of Natural History

Galactic Archaeology · **Invited Speaker**, SBCC Astronomy Club

Leadership & Outreach

2025 Committee Lead · *Commencement A/V Committee*, Earth & Planetary Sciences, UCSC

2023 - 2025 Member · *Astrobiology Reading Group*, Earth & Planetary Sciences, UCSC

Member · *Cosmochemistry Reading Group*, UCSC

2023 - 2024 Events Coordinator · *Women in Physics and Astronomy*, UCSC

2021 - 2023 President, Officer · *Astronomy Club*, Earth & Planetary Sciences, SBCC

2019 - 2023 Organizer and observatory operator · *Montly Star Party*, SBMNH, Santa Barbara, CA

2021 - 2023 Organizer and communicator · *Astronomy Club Star Parties*, Earth & Planetary Sciences, SBCC

2022 - 2023 Organizer and communicator · *STEM Day*, Earth & Planetary Sciences, SBCC

2023 Communicator · *Science Discovery Day (STEM Transfer Program)*, Earth & Planetary Sciences, SBCC

2022 Organizer and events host · *Astronomy Club Invited Talks, Trivia Night, etc.*, Earth & Planetary Sciences, SBCC

Communicator · *Ask an Astronomer Table*, Carnegie Observatories, Pasadena, CA

Communicator · *AstroFest 22 (Carnegie Observatories)*, Pasadena, CA

Communicator · *La Cumbre Junior High School Science Night (SBCC Astronomy Club)*, Santa Barbara, CA

Mentoring

2023-2026 Evan Grant · *Undergraduate*, UC Berkeley

2023-2026 CASSI Mentorship Program · *Undergraduates*, Carnegie Observatories Summer Student Internship

Skills

Leadership Event organizing and hosting; leading educational programs for school groups and community members; coordinating outreach; mentorship experience

Coding Python, R, LaTeX, DigiStar

Software COMSOL Multiphysics, iSALE, QGIS, ArcMap, Jupyter Notebook, Microsoft Suite, Open Broadcast Software, Moog, Spectroscopy Made Hard(er), StarFit

Languages Greek (native), English (native), French (elementary)